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Application of Fe(III)-exchanged clinoptilolite/graphite nanocomposite for electrochemical sensing of amitriptyline

Maria Madej[1,*], Katarzyna Fendrych[2], Radosław Porada[2], Mateusz Flacha[1], Jolanta Kochana[1], Bogusław Baś[2]

[1] Jagiellonian University, Faculty of Chemistry, Department of Analytical Chemistry, Gronostajowa 2, 30-387, Kraków, Poland

[2] AGH University of Science and Technology, Faculty of Materials and Ceramics, Department of Analytical Chemistry, A. Mickiewicza 30, 30-059, Kraków, Poland

*** Corresponding author:** Maria Madej, contact address: Faculty of Chemistry, Jagiellonian University, Gronostajowa 2, 30-387 Kraków, Poland; tel. + 48 126862399; e-mail: marysia.madej@doctoral.uj.edu.pl; https://orcid.org/0000-0002-7979-2922.

Abstract

This paper proposes a novel, ecofriendly voltammetric sensor for sensitive and selective amitriptyline (AMT) determination, based on glassy carbon electrode (GCE) surface modification by Fe^{3+} exchanged zeolite – clinoptilolite (FeZ) and graphite (G). The morphology and electrochemical properties of FeZ-G/GCE were investigated by X-ray powder diffraction (XRD), energy dispersion spectroscopy (EDS), scanning electron microscopy (SEM), electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV). The developed sensor showed improved catalytic activity and remarkable current increase for both model redox probe $[Fe(CN)_6]^{3-\backslash 4-}$ and AMT, in comparison to bare GCE and graphite modified G/GCE. In the analytical investigation of AMT using differential pulse voltammetry (DPV), the FeZ-G/GCE exhibited linear response in concentration ranges of 0.5 – 50.0 and 50.0 – 250.0 μmol L^{-1} with detection limit of 0.22 μmol L^{-1}. The proposed sensor was successfully applied for the determination of amitriptyline in pharmaceutical, environmental water and urine samples with recovery of 95.5 – 99.2%.

Keywords

amitriptyline, electrochemical sensor, pharmaceuticals, voltammetry, zeolite

1. Introduction

Depression is a serious, recurrent mood disorder that affects more than 320 million people worldwide and significantly impairs an individual's ability to function and cope with daily life. According to World Health Organization, depression is ranked as the largest contributor to global disability and major factor in suicide deaths, with almost 800 000 cases per year [1]. One of the most commonly used tricyclic antidepressant in clinical treatment of depression is 3-(10,11-dihydro-5H-dibenzo[a,d]cycloheptene-5-ylidene)-N,N-dimethylpropan-1-amine, known also under the name of amitriptyline (AMT). The mechanism of its action is based on inhibiting the reuptake of noradrenaline and serotonin from synaptic cleft to neurons, thus potentiating the activity of these neurotransmitters. Simultaneously, an amitriptyline blocks activated sodium channels and therefore has the ability to inhibit pain-generating ectopic activity in the central nervous system. Thanks to its anxiolytic, sedative and analgesic effects, amitriptyline is used not only as antidepressant drug, but also in the migraine or cancer treatment. AMT is metabolized in the liver with high efficiency - only 2% of the drug is excreted in unchanged form with urine. Its metabolite, nortriptyline, is a biologically active substance

with a similar mechanism of action, thus it also found application in the treatment of depression [2].

Because of inefficient removal of most pharmaceutical residues by wastewater treatment plants (WWTPs), low concentrations of various drugs, including AMT, remain widely distributed in aquatic ecosystems. The studies conducted over the past twelve years confirmed the presence of amitriptyline in diverse environmental media, such as river water, sewages or effluents of WWTPs in concentration ranges from 3.7 to 365 ng/L in countries like China, USA, Canada, France, Great Britain or Saudi Arabia [3-8]. Although the environmental concentration of AMT is relatively low, even at ng/L levels it may disturb the basic vital functions of various organisms like zebrafish or rodents [9-12]. These observations indicate that there is a great need to develop analytical methods, which would allow for precise and accurate determination of AMT in both biological and environmental samples.

Previously published methods of AMT quantification in commercial formulations, biological and water samples are mainly based on the use of separation techniques, including high performance liquid chromatography combined with spectrophotometric detection [13,14] or mass spectrometry [15,16], gas chromatography with mass spectrometry [17,18] and capillary electrophoresis combined with spectrophotometric detection [19] or time of flight detector [20]. Despite many advantages, separation techniques are limited by high detection costs, long analysis time, high consumption of solvents and the frequent need for complicated sample pretreatment including derivatization and extraction [21]. The application of electrochemical techniques allows to avoid these limitations, due to short response time, low cost of the apparatus combined with high sensitivity and good selectivity, even in the case of samples with complex matrices [22-25]. Few electrochemical methods of AMT determination have been reported in the literature so far. The AMT behavior was investigated using ion selective electrodes based on AMT-molybdovanadate/ molybdotungstate and AMT-phosphotungstate/ phosphomolybdate ion pair complexes [26,27]. For voltammetric determination, mainly carbon paste electrodes consisting of graphite powder mixed with carbon nanotubes [28], ionic liquid with montmorillonite nanoclay [29] or silica modified with niobium oxide, alumina and DNA [30] were employed. As well, sensors based on glassy carbon electrodes modification with nanomagnetic particles of Fe_3O_4 decorated with carbon quantum dots [31] or composite of multi-walled carbon nanotubes and cellulose [32] were reported. More complex nanomaterials, such as Nafion-AuNPs@ branched polyethyleneimine-derived carbon spheres immobilized at gold electrode surface were also employed for AMT determination [33].

High sensitivity, with respect to electrochemical sensors, is usually associated with high active surface area or good conductivity of working electrode, what can be achieved by the use of various nanomaterials for its modification e.g. carbon-based structures, conducting polymers, metal or metal oxide nanoparticles and materials with growing interest in that field like zeolites or metal-organic frameworks [34-36]. Zeolites constitute a large group of naturally occurring or synthetic crystalline, hydrated aluminosilicates characterized by numerous unique and specific physicochemical and structural properties, extremely relevant from the point of view of designing electrochemical sensors [35]. Zeolites framework composed of three-dimensional system of cages and channels with differential shapes but well-defined dimensions allows for selective adsorption of ions and molecules, what is responsible for their molecular sieving properties [37]. The presence of extra-framework cations balancing negative charge of the zeolite crystal lattice and having considerable freedom of movement, determine ion-exchange properties of these alluminosilicates. Thanks to these properties, there is a possibility for incorporation of various transition metals into zeolites structure towards exploiting them in electrocatalytic purpose [38]. Intensive use of zeolites in the modification of electrodes is

directly connected with mentioned-above microporous structure and high cation exchange capacity (CEC), as well as hydrophilicity, high thermal stability, insolubility in most organic solvents, and resistance in acidic media. Such wide range of advantageous features of zeolite modifiers resulted in growing interest of zeolite modified electrodes (ZMEs) in last decades and a number of reviews regarding preparation methods of modified electrode and discussion of electron-transfer mechanism have appeared in the literature [39-41]. Analytical applications of ZMEs in the determination of various analytes, including inorganic, organic and pharmaceutical compounds were also reported [42-44]. The vast majority of recently developed zeolite-based electrochemical sensors refers to fabrication of paste electrodes as a mixture of zeolite and conducting material (usually graphite) dispersed in epoxy resins or mineral oils [38,42,43]. Sensors prepared by this method often suffer from poor reproducibility, due to inhomogeneity in the resulting paste material [35].

In this work, we demonstrated a novel, environmentally friendly voltammetric sensor based on natural zeolite, clinoptilolite exchanged with Fe^{3+} ions for the determination of antidepressant drug, amitriptyline in neutral media. In contrast to commonly used preparation methods of zeolite-based sensors (focused on the preparation of paste electrodes), our sensor was prepared by modification of glassy carbon electrode surface with Fe^{3+}-zeolite-graphite mixture dispersed in polystyrene and acetone suspension. The morphological characterization of fabricated sensor was carried out using energy dispersion spectroscopy (EDS) and scanning electron microscopy (SEM). The electrochemical performance of Fe-zeolite/graphite modified glassy carbon electrode (FeZ-G/GCE) was verified using electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV) for model redox probe $[Fe(CN)_6]^{3-\backslash 4-}$ and AMT. Furthermore, the optimal conditions of analysis were selected and analytical parameters such as linear range, detection limit, short- and long-term stability were evaluated. Selectivity of fabricated sensor was verified by examining the influence of substances that usually occur in pharmaceutical, environmental and biological samples. Moreover, the usefulness of fabricated sensor was examined by determination of AMT in commercially available tablets, water, and urine samples.

2. Experimental

2.1. Apparatus

The voltammetric measurements were carried out using the M161 electrochemical analyzer (mtm-anko, Poland) connected to the electrode stand type M164 equipped with a magnetic stirrer. The EALab 2.1 software was applied for data acquisition. Experiments were conducted in three-electrode system with bare or modified glassy carbon electrode (GCE) (MF-2012, φ = 3 mm, BASi, USA) as a working electrode, silver chloride electrode (Ag/AgCl, 3M KCl, Mineral, Poland) as a reference electrode and platinum wire placed in Teflon® holder as an auxiliary electrode. Electrodes were placed in measurement cell of 15 mL volume covered with a Teflon® cover with four holes, that matched the size of each electrode, and an additional hole for adding standard solution. The impedance spectra were recorded with the use of μAUTOLAB III analyzer (EcoChemie, the Netherlands) with NOVA 2.0 software.

The phase composition of zeolite material was determined by means of a X-ray powder diffraction X'Pert system (CuKα) (Philips, the Netherlands). The XRD pattern was obtained in the range 5-80° 2θ, with a step of 0.016°. The X'Pert HighScore Plus software and the International Centre for Diffraction Data (ICDD) cards were used for data analysis. Rietveld's method was used to establish the quantitative phase composition.

Energy-dispersive X-ray spectroscopy EDS attachment of scanning electron microscope FEI Nova NanoSEM 200 (ThermoFisher Scientific, USA) was used for the elemental analysis of initial zeolite material, as well as the zeolite after ion-exchange modification with Fe^{3+} cations. Before measurement in vacuum, the samples were sprayed with graphite.

Buffer solutions were prepared using pH-meter CP-511 with combined glass electrode EPS-1 (Elmetron, Poland). Ultrasonic bath Sonic-3 (Polsonic, Poland), Vortex mixer (Labnet International Inc., USA), magnetic stirrer (Wigo, Poland), laboratory incubator ST1 (Pol-eko, Poland) and Quintix® Analytical Balance (Sartorius, France) were also used.

2.2. Reagents and materials

Before surface modification, the working electrode was polished using MicroPolish Alumina suspension with grain size 0.05 μm (Buechler, USA) on polishing cloths (Leco, USA). Natural zeolite material collected in Transcarpathian Zeolite Plant (Ukraine) was purchased from the Andalusia Group. Graphite powder (< 20 μm) and polystyrene with average molecular weight 192,000 were purchased from Merck (Germany). The standard stock solution of 10.0 mmol L^{-1} amitriptyline hydrochloride (Merck, Germany) was obtained by dissolving of an appropriate amount of drug in 2 mL of water and then stored in the fridge. Nitric acid (65%), potassium chloride, sodium hydroxide, acetone, disodium hydrogen phosphate, sodium dihydrogen phosphate and ascorbic acid were purchased from Avantor Performance Materials Poland S.A. (Poland). Lithium chloride, iron(III) chloride, boric acid, phosphoric acid, potassium nitrite, potassium hexacyanoferrate(II) and (III) were obtained from Chempur (Poland). Sodium chloride and hydrochloric acid (35%) were purchased from Lachner (Czech Republic). Ammonia, ammonium chloride and sulphuric acid were obtained from Merck (Germany). The Britton-Robinson buffers at pH 2 and 4 were prepared by mixing appropriate volumes of 0.2 mol L^{-1} sodium hydroxide solution and 0.04 mol L^{-1} mixture of phosphoric acid, boric acid and acetate acid. The phosphate buffers (pH 6 and 8) were prepared by mixing appropriate volumes of 0.1 mol L^{-1} disodium hydrogen phosphate solution and 0.1 mol L^{-1} sodium dihydrogen phosphate solution. The 0.1 mol L^{-1} ammonium buffer at pH 10 was prepared by mixing appropriate volumes of ammonia and 0.1 mol L^{-1} ammonium chloride. For interference studies ascorbic acid, glucose (Microparm, Poland), lactose, starch, talc, titanium dioxide, Triton X-100, humic acid, sodium dodecyl sulphate, magnesium stearate, cetrimonium bromide (all purchased from Sigma-Aldrich, USA), certified reference material of synthetic urine Seronorm Trace Elements Urine (Sero, Norway) and certified reference material of synthetic serum Nortrol (Thermo Scientific, Finland) were used. For the verification of method applicability, drug containing amitriptyline hydrochloride Amitryptylinum VP (Valeant Pharmaceuticals, Canada) was purchased from local pharmacy and water sample from Vistula River was collected in Krakow. The Certified Reference Materials of Surface Water SPS–SW1 Batch no. 114 and Waste Water (SPS–WW2 Batch no. 113 (Spectrapure Standards AS, Norway) were used to investigate the usefulness of developed method for analysis of samples with complex matrix. All aqueous solutions were prepared with chemicals of analytical grade purity and distilled water from the HLP 5 system (Hydrolab, Poland).

2.3. Standard procedures

2.3.1. Zeolite modification and zeolite-based sensor fabrication

The modification of zeolite with Fe^{3+} cations, as well as electrode preparation was performed according to previously reported procedure [44]. To ensured complete ion-exchange process, and thus to receive Fe^{3+} form of zeolite, 0.5 g of zeolite sample was treated two times with 5 mL of 0.1 mol L^{-1} $FeCl_3$ aqueous solution, shaken for 24 h at 25°C, and centrifuged at 15 000

rpm for 10 min. The obtained Fe^{3+} form of zeolite was washed three times with a small volume of double-distilled water, dried at room temperature and used for the surface modification of glassy carbon electrode. For that purpose, 20 mg of ground mixture (1:1) of Fe-zeolite and graphite was weighted and mixed with 500 µL of acetone solution containing polystyrene at concentration 20 mg mL^{-1}. Further, the resulting suspension was homogenized using ultrasonic bath for 15 minutes and then stored in the fridge until use.

Prior to surface modification, the GCE electrode was polished to mirror-like surface on a polishing cloth with a drop of alumina slurry for 30 seconds, then rinsed with distilled water and sonicated for 5 minutes in a mixture of water and acetone (volume ratio 1:1). Afterwards, the electrode was dried at room temperature, then 20 µL of zeolite-graphite suspension, previously mixed for 5 min using vortex, was casted onto the surface of the pretreated GCE and left to dry in laboratory incubator at temperature of 22°C for 24 hours. All sensors were stored under the above conditions also between measurements. For measurements conducted at low concentrations of amitriptyline the incubation time was extended to 14 days in order to minimize the background current.

2.3.2. Measurement procedure

Preliminary electrochemical measurements related to examination of the sensor performance in a model redox probe $[Fe(CN)_6]^{3-\backslash 4-}$ were conducted using cyclic voltammetry in the potential range between -0.4 to 0.8 V at scan rate of 0.05 V s^{-1}. The impedance spectra were recorded at the formal potential of the redox pair $[Fe(CN)_6]^{3-\backslash 4-}$ using sinusoidal signals with frequency in the range of 100 kHz – 25 mHz and amplitude of 10 mV in 0.1 mol L^{-1} KCl containing 0.2 mmol L^{-1} concentration of $[Fe(CN)_6]^{3-\backslash 4-}$. In order to determine the quantitative parameters characterizing the processes of charge transfer, mass transport and charging the electrical double layer, an equivalent circuit was used, the spectrum of which was adjusted to the experimental spectrum. Measurements were carried out in a deoxygenated solution.

Electrochemical behavior of AMT on FeZ-G/GCE was tested in 0.1 mol L^{-1} potassium chloride solution by cyclic voltammetry in the potential range from 0 to 1.4 V with scan rate equal 0.05 V s^{-1}. Before measurements, the content of the measuring cell was stirred (ca. 500 rpm) for 5 min without applying any potential to the working electrode.

AMT determination was carried in 0.5 mol L^{-1} lithium chloride solution as a supporting electrolyte using differential pulse voltammetry in the anodic direction from 0 to 1.45 V with potential step (E_s) 7 mV, pulse width (t_p) 20 ms (10 ms of waiting time and 10 ms for current sampling time) and pulse amplitude (dE) 50 mV. Before each measurement, accumulation potential (E_{acc}) of 0 mV for 3 min (t_{acc}) was applied to the working electrode, while the solution in measurement vessel was being stirred (ca. 500 rpm). Measurements were repeated three times and for data analysis, the mean value was taken into consideration. For each mean current value, the confidence interval (CI) was calculated based on Equation 1 and included at corresponding graphs in a form of error bars:

$$CI = t_{\alpha;r} \cdot \frac{\sigma}{\sqrt{n}} \quad \text{(Eq.1)}$$

where: $t_{\alpha;r}$ – critical value of Student's t distribution for significance level $\alpha = 0.05$ and degrees of freedom r = 2, σ – standard deviation, n – number of measurements.

2.3.3. Samples preparation

Pharmaceutical containing 10 mg of AMT per one tablet, Amitryptylinum VP was purchased in local pharmacy. The stock solution with expected AMT concentration of 1.275 mmol L^{-1} was prepared. At first, 3 tablets of tested pharmaceutical were weighted and the average weight of one tablet has been determined. Then, all tablets were ground in an agate mortar, appropriate portion of obtained powder was weighted and dissolved in 50 mL of distilled water followed by sonication for 15 min. Afterwards, prepared solution was subjected to filtration. For quantitative determination of AMT, 80 µL of stock solution was added to 10 mL of supporting electrolyte and then measurements were conducted according to the standard addition calibration method. For each concentration, three subsequent voltammograms were recorded, and after background subtraction, the mean value of achieved peak current values was calculated.

The sample of river water was taken from the Vistula River in Krakow, subjected to filtration and stored in the fridge before analysis. River water samples and certified reference materials of surface water and waste water were enriched with appropriate volume of 10 mmol L^{-1} standard stock solution of amitriptyline hydrochloride, to obtain concentrations of 20.0 µmol L^{-1}. For DPV determination, 2.5 mL of spiked samples were added to the 7.5 mL of supporting electrolyte and measurements were conducted similarly as for pharmaceutical.

The synthetic urine certified reference material was spiked with 10 mmol L^{-1} AMT standard stock solution in order to obtain concentration of 250 µmol L^{-1}. For quantitative determination, 200 µL of spiked urine sample was added to the 9.8 mL of lithium chloride solution and then measurements were performed as described for previous samples.

3. Results and discussion

3.1 Composite characterization

In this work, volcanic-sedimentary origin zeolite material from Sokirnyanskyy deposit (Ukraine) was utilized. The mineralogical composition, determined based on registered XRD pattern and Rietveld's method, revealed that the used material consisted of 65.1% ca-clinoptilolite, 20.5% of quartz and 14.4% of muscovite [44]. Clinoptilolite, belongs to heulandite group, and it is one of the most abundance and the most often used natural zeolite among more than 40 known of natural alluminosilicates.

The incorporation of Fe^{3+} cations into zeolite structure was confirmed by use of energy-dispersive X- ray spectroscopy EDS measurements. The analysis of EDS spectrum of initial natural zeolite material and zeolite after ion-exchange modification with Fe^{3+} (see Fig. S1 in supplementary data) showed, that peaks related to atoms building zeolite framework (Si, Al, and O) and exchangeable cations (Na^+, K^+, Mg^{2+} and Ca^{2+}) were recorded. Moreover, in both cases the EDS spectrums indicated also the presence of Fe cations in zeolite structure, the applied procedure of zeolite modification caused an increase of the iron content, expressed as the average value of the atomic concentration of several, different surface points, from 0.76 to 1.92 at.% for initial zeolite material and Fe-zeolite, respectively.

3.2 Electrochemical characterization of Fe-zeolite based sensors

Electrochemical impedance spectroscopy was employed to examine electrochemical properties of GCE, graphite-modified GCE and Fe-zeolite/graphite modified GCE in 0.1 mol L^{-1} KCl solution as supporting electrolyte in the presence of $[Fe(CN)_6]^{3-/4-}$ with a concentration of 0.2 mmol L^{-1} as model redox probe. The recorded impedance spectra for all tested electrodes in the form of a Nyquist plot are shown at Fig. 1a. The dashed line marks the characteristic

semicircle representing the charge transfer resistance and charging of the electric double layer, obtained as a result of fitting the model spectrum to the high-frequency fragment of the spectrum. The values of charge transfer resistance (R_{ct}), Warburg impedance module (σ) and differential capacity of the double layer (C_{dl}) obtained as a result of fitting the model spectra, as well as effective electrode surface (A_{eff}) and heterogeneous rate constant of the electrode reaction (k_s) determined on their basis are presented in Table 1.

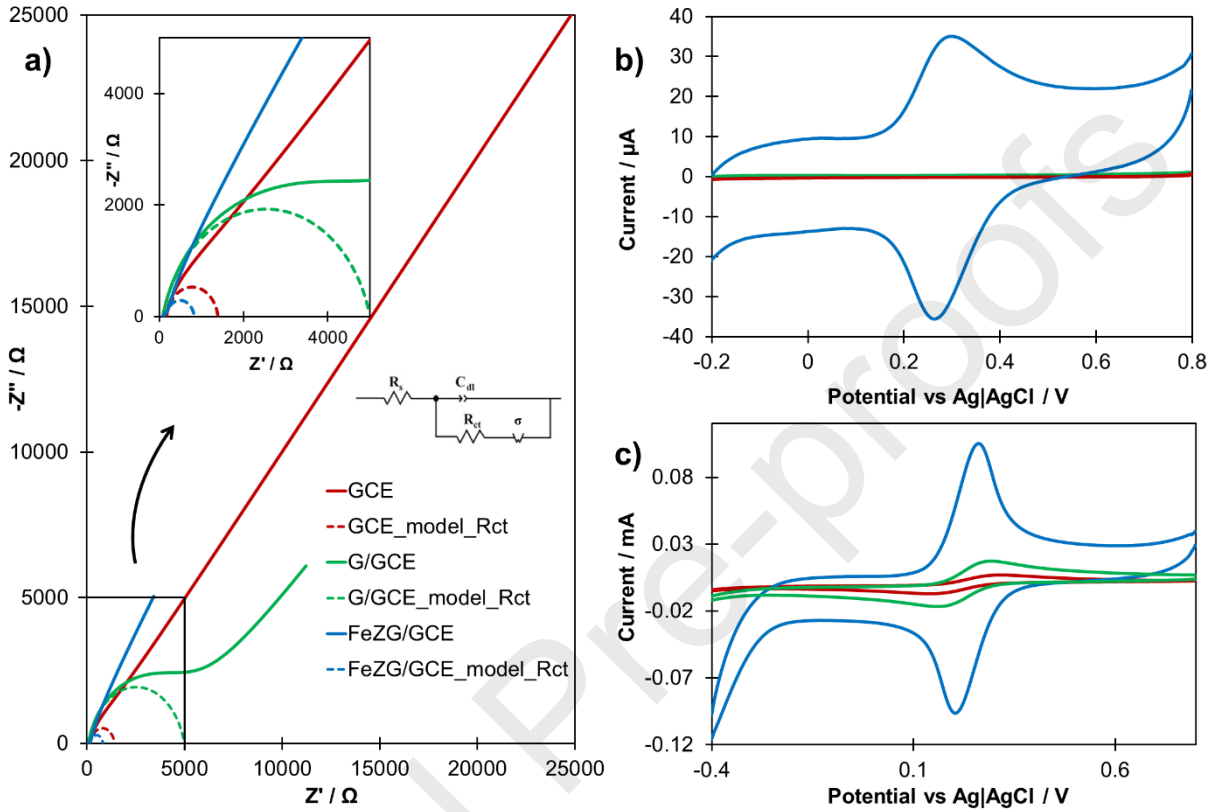


Figure 1. The impedance spectra (a) recorded for $[Fe(CN)_6]^{3-\backslash 4-}$ (0.2 mmol L^{-1}) in 0.1 mol L^{-1} KCl solution and cyclic voltammograms recorded without (b) or in the presence of 1.0 mmol L^{-1} $[Fe(CN)_6]^{3-\backslash 4-}$ (c) using GCE (red line), graphite/GCE (green line) and Fe-zeolite-graphite/GCE (blue line).

Table 1. The comparison of tested electrodes based on parameters achieved from EIS measurements (n = 3).

Parameter	Working electrode		
	GCE	G/GCE	FeZ-G/GCE
R_{ct} / kΩ	1.23 ± 0.01	4.92 ± 0.07	0.65 ± 0.01
σ / kΩ $s^{-1/2}$	22.3 ± 0.1	5.1 ± 0.2	2.0 ± 0.1
C_{dl} / µF cm^{-2}	28.9 ± 0.3	2.4 ± 0.1	1154 ± 50
A_{eff} / mm^2	3.13 ± 0.02	14.3 ± 0.2	35.7 ± 0.4
k_s / m/s	(3.47 ± 0.03) · 10^{-4}	(1.93 ± 0.08) · 10^{-6}	(5.8 ± 0.1) · 10^{-5}

Recorded EIS spectra and parameters submitted in Table 1 prove that tested electrodes exhibit different behavior. In the case of the unmodified GCE, the Nyquist plot primarily shows a straight line inclined at 45°, which corresponds to the Warburg impedance associated with the diffusive transport of the depolarizer to the electrode surface. The fragment responsible for the charge transfer resistance is visible only for very high frequencies (above 1 kHz). Due to the fact that the Warburg impedance modulus is approximately 20 times greater than R_{ct} it could be assumed that occurring electrode reaction is a reversible process. The obtained value of differential capacity is typical for 0.1 mol L^{-1} KCl, which proves that no additional adsorption or ion exchange processes occurs at GCE. The determined real surface coincides with the geometric surface of the electrode (3.1 mm^2).

Among tested electrodes, the GCE modified with graphite is distinguished by a very high R_{ct} value, which indicates that the presence of graphite-polystyrene composite at the surface of GCE hinders the charge transfer process. The development of the surface compared to the unmodified electrode contributes to a lower value of the Warburg impedance modulus, thus the diffusion process is relatively easy and, in combination with charge transfer, it could be defined as a reaction rate-limiting step (quasi-reversible process).

The introduction of natural zeolite modified with Fe^{3+} ions to the modifying layer containing graphite resulted in an 8-fold decrease in the R_{ct} value, which indicates both faster electron transfer kinetics and a significant expansion of the electrode surface (approx. 10 times compared to the GCE and approx. 2.5 times in comparison to G/GCE). As in case of G/GCE, the increase in surface area of FeZ-G/GCE also translates into a decrease in the value of the Warburg impedance modulus. The comparable values of R_{ct} and σ indicate that occurring electrode reaction is a quasi-reversible process. The high value of differential capacity is directly related to the presence of natural zeolite, which has very good sorption properties and ion exchange capacity. It is also worth noting that in the tested frequency range, the real and imaginary part of the impedance of the FeZ-G/GCE electrode do not exceed 5 kΩ, while for the remaining electrodes values of 40 kΩ and greater are achieved, which translates into greater ohmic drops for these electrodes.

The CV measurements conducted with the use of all tested electrodes in 0.1 mol L^{-1} KCl solution without (Fig. 1b) and in the presence of 1.0 mmol L^{-1} $[Fe(CN)_6]^{3-\backslash 4-}$ (Fig. 1c) were also recorded. As can be seen, no voltammetric peaks were observed in KCl solution for bare GCE and graphite-modified GCE, which proves that supporting electrolyte solution did not contain any electroactive species and graphite layer deposited on the surface of GCE is also electrochemically inert. In opposite, the GCE modified with a Fe-zeolite and graphite showed well-defined peaks associated with occurring oxidation and reduction of iron cations. These observations suggest that in the presence of supporting electrolyte Fe^{3+} cations from zeolite pores are exchanged with K^+ cations from the solution and then undergo oxidation/reduction processes at the electrode surface, in accordance with the extrazeolitic electron transfer process [41].

The cyclic voltammograms recorded in the presence of model redox probe (Fig. 1c) using FeZ-G/GCE shows a remarkable enhancement of the peak current (approx. 17 times compared to the GCE and approx. 7 times in comparison to G/GCE). For tested electrodes, peak current value can be described by the ratio 1 : 2.7 : 14.5, while the surface ratio equals 1 : 4.6 : 11.4 for GCE, G/GCE and FeZ-G/GCE, respectively. As can be seen from Ilkovic equation, in the same conditions, for the same concentration of the depolarizer, the current is proportional to the electrode surface. Therefore, for the electrode with a higher surface, higher peak currents should be registered. The fact that the peak current for the FeZ-G/GCE is higher than expected based

on the surface ratio can be associated with the catalytic properties of the zeolite. For FeZ-G/GCE the difference between the anodic and cathodic peak potential (ΔE) is about 56.2 mV, which indicates occurring a reversible electron transfer, while for G/GCE and GCE the ΔE is equal 140 mV and 85 mV, respectively. Moreover, the shift in the oxidation-reduction peaks potentials to lower values was also observed. These results proves that application of Fe^{3+} exchanged zeolite for surface modification facilitates the charge transfer and enhance the electrocatalytic properties of developed sensor in comparison to GCE and G/GCE.

3.3 Electrochemical behavior of amitriptyline at FeZ-G/GCE

The electrochemical behavior of AMT has been evaluated through CV by using GCE, graphite-GCE and Fe-zeolite/graphite modified GCE in 0.1 mol L^{-1} solution of potassium chloride containing 50 μmol L^{-1} of AMT (Fig. S2). Recorded voltammograms obtained using all tested electrodes proved that AMT undergoes an irreversible oxidation process with one anodic wave. The bare GCE showed a poor anodic current (1.21 μA) with peak potential at 1.34 V. After modification of GCE with graphite, the oxidation peak current decreased to 0.66 μA with peak potential at 1.33 V. In a case of FeZ-G/GCE the peak potential shifted towards a negative values (1.28 V) with remarkable, more than fifteen-fold increase of recorded peak current (20.01 μA). It is clearly visible that significant improvement of sensitivity and electrocatalytic activity is associated with the presence of Fe^{3+} modified zeolite in tested composite, which contributes to obtaining a large surface area and facilitating the electron transfer. The enhancement of charge transfer properties has also been confirmed in EIS and CV measurements conducted for redox probe $[Fe(CN)_6]^{3-\backslash 4-}$ solution (see Section *3.2 Electrochemical characterization of Fe-zeolite based sensors)*.

According to previously reported studies related to electrooxidation of various alkylamines, it can be assumed that observed anodic peak is related to oxidation of tertiary amine group, which leads to the formation of a cation radical [45,46]. In order to verify this thesis, the effect of scan rate (ν) on the oxidation peak current (I_p) was investigated at FeZ-G/GCE in the range from 12.5 to 500 mV s^{-1} by CV. The voltammograms presented at Fig. 2a shows that with increasing scan rate the oxidation peak current also increased. Furthermore, the peak potential shifted towards more positive values, which confirms the irreversibility of AMT oxidation process [47]. An excellent linearity between peak current and $\nu^{1/2}$ was obtained with a regression equation of I_p (μA) = 3.901 $\nu^{1/2}$ – 0.807 (R^2 = 0.9927, see Fig. S3 in supplementary data). In addition, a dependence of logarithm of peak current on the logarithm of scan rate (Fig. 2b) is characterized by good linearity (R^2 = 0.9840) with directional coefficient of 0.5137, which is close to theoretical value of 0.5, expected for diffusion-controlled process [47].

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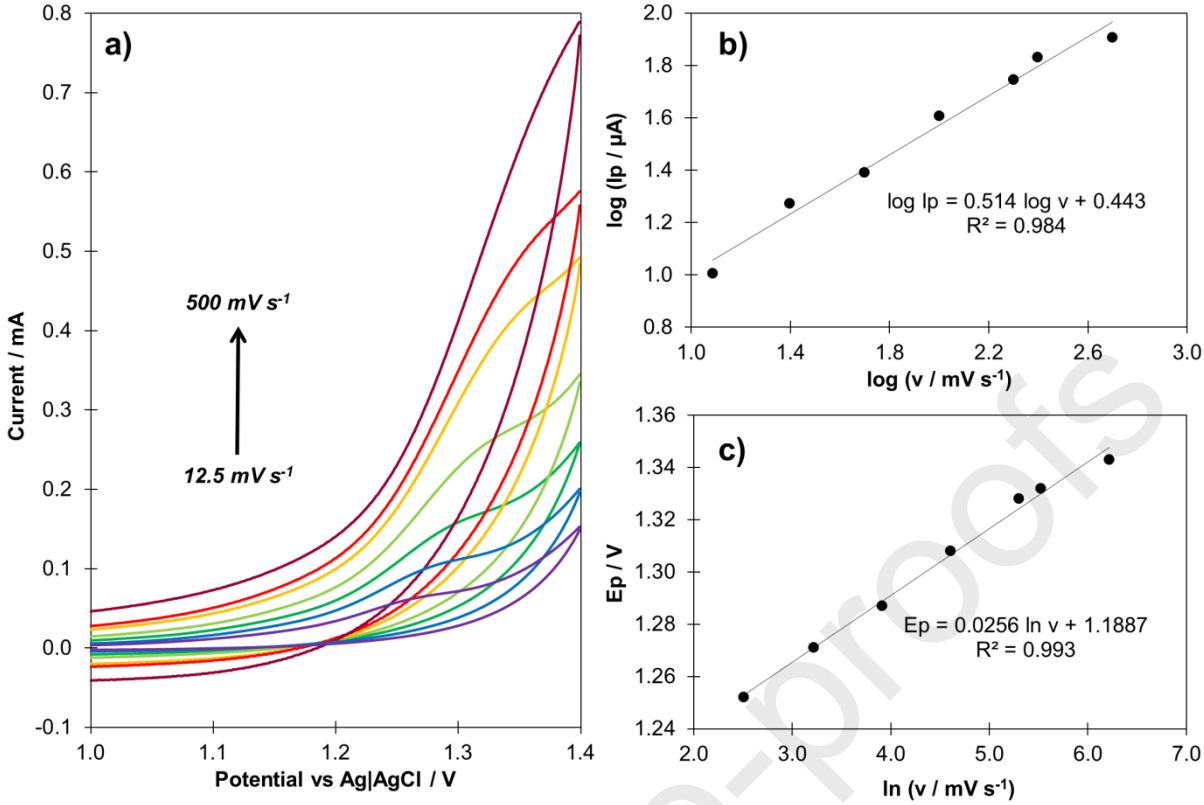


Figure 2. Cyclic voltammograms registered for increasing scan rate values (a) and their influence on AMT oxidation peak current (b) and peak potential (c) performed in 0.1 mol L^{-1} KCl solution containing 50 µmol L^{-1} AMT.

As can be seen from recorded voltammograms (Fig. 2a), scan rate had influence not only on AMT oxidation peak current, but also on the peak position - oxidation peak potential shifted towards positive values with increasing scan rate. For a typical irreversible, diffusion-controlled process relation between oxidation peak potential (E_p) and scan rate can be described by following equation [48]:

$$E_p = A + \frac{RT}{2\alpha nF} ln\ v \quad \text{(Eq.2)}$$

where: E_p – peak potential [V], A – constant value, R – gas constant (R = 8.314 J mol^{-1} K^{-1}), T – temperature [K], α – charge transfer coefficient, n – number of electrons, F – Faraday constant (F = 96 485 C mol^{-1}), ν – scan rate [mV s^{-1}]. Based on the plot of E_p versus ln ν presented at Fig. 2c, number of electrons involved in the rate-determining step was calculated. Assuming standard value of charge transfer coefficient for irreversible process α = 0.5, number of electrons was calculated as 1.003, which confirmed one-electron transfer for AMT oxidation process.

The effect of pH on the electrochemical response of FeZ-G/GCE in the presence of AMT was investigated over the pH range 4 - 10 using CV. The voltammograms were recorded in different types of buffers with the same concentration of 0.1 mol L^{-1} – Britton-Robinson (B-RBS), phosphate (PBS) and ammonium (ABS) buffers, containing 50 µmol L^{-1} of AMT, and the influence of pH value on Ep and Ip was examined (Fig. S4). It was observed that the oxidation peak potential shifted towards negative direction with increasing pH, which confirms the participation of proton in the electrode process. Moreover, for the dependence between peak

potential and pH (Fig. S4b) a slope of -58.9 mV pH^{-1} was obtained, which suggest that equal number of protons and electrons are involved in the electrode reaction [49].

Based on the conducted research, it can be concluded that electrode reaction proceeds via irreversible oxidation of tertiary amine group in amitriptyline molecule through the creation of an aminium cation radical. The occurring process is accompanied by the release of one electron and one proton. Suggested mechanism is consistent with literature reports associated with AMT oxidation investigations conducted at different sensors [28, 33]. The proposed mechanism of electrode reaction is presented at Fig. S5.

3.4 Optimization of experimental parameters

3.4.1 Choice of supporting electrolyte

Due to specific pore size and cavity dimensions of each zeolite, ion exchange process depends distinctly on the type and concentration of supporting electrolyte, which also determines the electrochemical performance of zeolite-based sensors [38]. Therefore, the influence of previously tested buffer solutions on the height of the recorded signal was investigated. Voltammograms presented at Fig. S4a clearly shows that the AMT oxidation peak current, increased with a growth of buffer pH, reached maximum at pH 6, and then slightly decreased. Nevertheless, the peak current obtained in phosphate buffer at pH 6 was distinctly lower than peak current recorded in 0.1 mol L^{-1} potassium chloride solution used as supporting electrolyte at preliminary studies (see Fig. S2 (inset) and Fig. S4a). Hence, different strong electrolytes such as NaCl, LiCl, HCl, H_2SO_4 and KNO_3 were examined as possible supporting electrolytes. For that purpose, the cyclic voltammograms were recorded in 0.1 mol L^{-1} solutions of mentioned salts or acids in the presence of AMT in concentration of 50 µmol L^{-1}. As can be seen from Fig. S6a, the highest oxidation peak current was obtained in lithium chloride solution, with approx. increase of 34% and 82% in comparison to potassium chloride and phosphate buffer solutions, respectively. For the other tested compounds, lower peak current values were observed, with no recorded signal in sulfuric acid solution. The best performance of FeZ-G/GCE in lithium chloride solution can be attributed to the smallest ionic radius of Li^+ compared to the size of cations present in other tested solutions. Small cations can easily enter the zeolite pores and effectively exchange with Fe^{3+} cations, which results in enhancement of electrocatalytic properties of zeolite, and thus promoting the AMT oxidation process. These observations are consistent with previous reports regarding ZMEs behavior in various supporting electrolytes [38,42]. In addition, the influence of LiCl concentration (in the range of 0.05 – 1.0 mol L^{-1}) on registered peak current was also verified. As can be seen from Fig. S6b, obtained AMT oxidation peak current increased with higher concentration of LiCl, reached maximum value for 0.5 mol L^{-1} and then declined evenly. Thus, 0.5 mol L^{-1} lithium chloride solution was chosen as suitable supporting electrolyte.

For previously described voltammetric sensors used for AMT determination strongly acidic medium was often required. Furthermore, the AMT oxidation peak current significantly decreased or disappeared completely in solutions with pH equal or higher than 4.0 [28, 31, 33]. Therefore, it is worth emphasizing that FeZ-G/GCE stands out from the above-mentioned sensors by exhibiting a wide pH range (2 – 10) at which AMT oxidation peak was observed, with best response achieved for environmentally neutral supporting electrolyte such as lithium chloride, which promotes the principles of green analytical chemistry [50].

3.4.2 Optimization of DPV parameters

DP voltammetry was chosen for quantitative determination of AMT due to higher sensitivity than CV and ability to reduce the impact of charging current. Thus, in order to improve the performance of developed sensor, the optimization of DPV parameters with previously chosen supporting electrolyte was conducted. At first, different values of potential step (E_s) in the range of 2 – 8 mV, with constant pulse width of 40 ms and pulse amplitude of 40 mV, were tested (Fig. S7a). Comparable values of peak current were obtained for potential step of 7 and 8 mV, however with higher E_s, the resolution of measurement decreases, therefore E_s = 7 mV was chosen for further studies. In the next stage, the potential step was kept constant at 7 mV with pulse amplitude of 40 mV, while the pulse width (t_p) was varied from 10 ms to 200 ms (Fig. S7b). The exponential growth of peak current with a decreasing pulse width was observed. However, for the shortest tested value (t_p = 10 ms), despite of high registered peak current, poor repeatability was achieved. Therefore, the pulse width equal 20 ms (10 ms of waiting time and 10 ms for current sampling time) was applied. Also different pulse amplitude (dE) values from 10 to 80 mV (both positive and negative modes) were tested, applying previously selected E_s and t_p values (Fig. S7c). The increase of peak current with the growth of dE, with simultaneous decrease of registered signals repeatability, was observed. The similar behaviour was noted for negative modes, but lower peak current values were achieved. Thus, the pulse amplitude equal 50 mV was selected. Finally, the possible influence of accumulation time (t_{acc}) on peak current was investigated. For that purpose, various time of preconcentration step was tested, applying previously optimized parameters and accumulation potential (E_{acc}) equal 0 mV (Fig. S8a). The recorded peak current increased distinctly with a growth of accumulation time up to 3 minutes, and then subsequent extension of time did not affect the peak current visibly. Thus, t_{acc} equal 3 min was selected as an optimal value. In order to improve sensitivity of FeZ-G/GCE, the application of various accumulation potential, with previously chosen accumulation time, was also tested (Fig. S8b). The growth of accumulation potential resulted in decrease of recorded peak current, which suggests that applying positive potentials to the working electrode reduces the efficiency of AMT accumulation on the surface of developed sensor. Therefore, the final DP measurements were preceded by applying to the modified electrode the accumulation potential equal 0 mV for 3 minutes.

The final procedure of AMT determination using DPV technique was performed in lithium chloride solution (0.5 mol L^{-1}) in the anodic direction from 0 to 1.45 V with potential step 8 mV, pulse width 20 ms and pulse amplitude 50 mV. Before each measurement, accumulation potential of 0 mV was applied to the working electrode for 3 min.

3.5 Analytical performance

DP voltammetric measurements were conducted under previously selected experimental conditions to investigate the analytical performance of Fe-zeolite/graphite modified GCE towards AMT determination. For that purpose, DP voltammograms were recorded for increasing concentrations of AMT in 0.5 mol L^{-1} LiCl solution (Fig. 3).

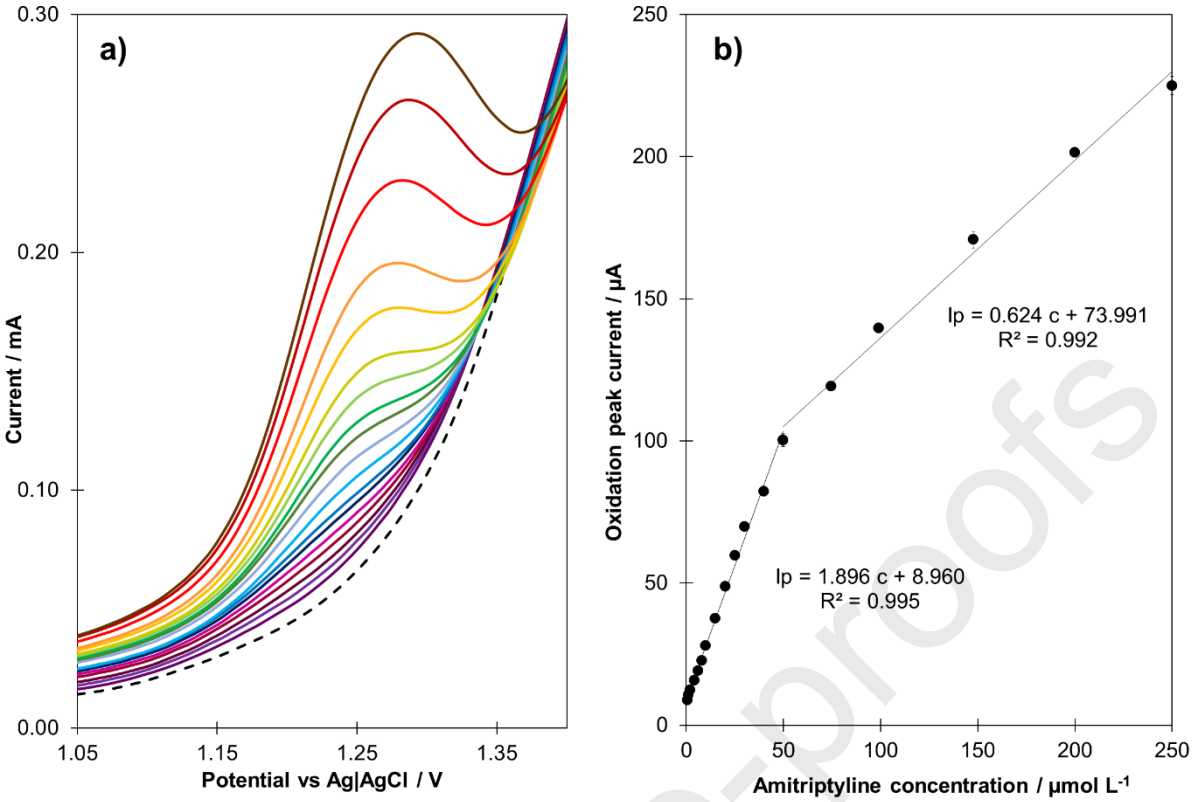


Figure 3. DP voltammograms (a) and corresponding calibration graph (b) achieved for different concentrations of AMT (solid lines) in lithium chloride solution (dotted line) at concentration of 0.5 mol L^{-1}. DPV parameters: E_s = 7 mV; t_p = 20 ms; dE = 50 mV; t_{acc} = 180 s; E_{acc} = 0 mV.

As can be seen, a good linear correlation between oxidation peak current and AMT concentration was found. The FeZ-G/GCE exhibited linear response in two concentration ranges: 0.5 – 50.0 and 50.0 – 250.0 µmol L^{-1}, with sensitivity of 1.77 ± 0.14 µA L µmol^{-1} and 0.66 ± 0.04 µA L µmol^{-1} (mean values of 3 sensors), respectively. The limit of detection, LOD = 0.22 µmol L^{-1}, was estimated as a 3.3 times standard deviation of peak current registered for 0.5 µmol L^{-1} AMT concentration calculated from five subsequent repetitions using one sensor over the mean value of calibration slope. The limit of quantification, LOQ = 0.67 µmol L^{-1}, was evaluated as 3LOD [51]. It is worth mentioning that oxidation peak current recorded for the same concentration of AMT at various sensors differed noticeably from each other. However, this divergence did not affect the sensitivity of the measurements, therefore reproducibility and repeatability were tasted on the basis of sensitivity obtained for lower linear range. The reproducibility was calculated as relative standard deviation of sensitivity values obtained using five different sensors prepared independently under the same conditions. While repeatability was evaluated based on five calibration sets obtained subsequently using the same sensor. The results display a satisfactory reproducibility of 7.7% and good repeatability of 2.4%. Additionally, the long-term stability was investigated by comparing the sensitivity of fabricated sensor after storage at 22°C for 3, 5, 7, 11, 14, 21, 30 and 60 days. The sensitivity did not fall below 95.2% of the initial value after 21 days, while after 30 and 60 days, sensitivity decreased to 93.1% and 90.7%, indicating excellent long-term stability of the developed sensor. Moreover, with the extension of the storage time, a decrease of background current was observed, and thus, in lower linear range the recorded peaks were better shaped. Therefore, for further studies conducted at low AMT concentrations, the storage time of 14 days was adopted. In comparison to previously reported electrochemical methods of AMT determination

(described in Table 2), fabricated Fe-zeolite/graphite modified glassy carbon electrode exhibit wide linear range and satisfactory LOD value.

Table 2. Comparison of electrochemical methods developed for AMT determination

Electrode	Technique	Linearity range / $\mu mol\ L^{-1}$	LOD / $\mu mol\ L^{-1}$	Sample	Ref.
$SiO_2/Al_2O_3/Nb_2O_5$/ DNA/CPE[1]	DPV	10 – 80	0.120	tablets	[30]
NC-CILE[2]	AdSV	0.1 – 8 8 - 50	0.024	tablets, serum	[29]
CNTE[3]	DPV	0.0 – 30	1.610	tablets	[28]
Nafion-AuNPs@ bPEI-CHS/AuE[4]	CV	0.1 – 700	0.033	tablets, serum	[33]
Amt-Mv/ISE[5]	potentiometry	100 - 10 000	68.7	tablets, water	[27]
MagNPs/Cdots/ GCE[6]	DPV	0.05 – 13.50	0.006	-	[31]
MWCNT@ Cellulose/GCE[7]	DPV	0.5 – 20.0	0.085	tablets, urine	[32]
FeZ-G/GCE	**DPV**	**0.5 –50.0 50.0 – 250.0**	**0.22**	**tablets, water, urine**	**This work**

[1] carbon paste electrode composed of graphite powder mixed with mineral oil and silica modified with niobium oxide, alumina and DNA; [2] carbon ionic liquid electrode with incorporated montmorillonite nanoclay; [3] carbon nanotube electrode; [4] gold electrode modified with gold nanoparticles branched polyethyleneimine-derived carbon hollow spheres coated with Nafion; [5] poly(vinyl chloride) membrane electrode based on amitriptyline-molybdovanadate ion pair complex; [6] Fe_3O_4 nanoparticles decorated with quantum dots immobilized onto glassy carbon electrode with neodymium magnet; [7] multi-walled carbon nanotubes-cellulose composite modified electrode.

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3.6 Interference studies

In order to evaluate the selectivity of fabricated sensor, the influence of different inorganic and organic compounds, which could be found in pharmaceuticals and environmental water samples, on the recorded signal was investigated. For interference studies ascorbic acid, glucose, lactose, magnesium stearate, talc, starch, titanium dioxide, Triton-X100 (non-ionic surfactant), sodium dodecyl sulphate (SDS; anionic surfactant), cetrimonium bromide (CTAB; cationic surfactant), humic acid and inorganic ions (Na^+, K^+, Cl^-, NO_3^-, SO_4^{2-}) were chosen. For each tested interferent (in concentration between 1.0 – 200 μmol L^{-1}) a percentage difference between peak current obtained for 50 μmol L^{-1} of AMT recorded without and in the presence of interferent was calculated. The acceptable signal change was defined as percentage difference less than ± 5%.

Performed measurements indicated that developed sensor is characterized by high selectivity. Among tested inorganic compounds, none of them affected the shape, current and potential of AMT oxidation peak. In the case of most of tested organic interferents, the signal change did not exceed the tolerance limit (for more details see Table S1). Only CTAB, Triton X-100 and magnesium stearate had negative impact on the height of recorded signal (Fig. 4a), without change of signal shape and its position. It is clearly visible that increasing concentration of non-ionic and cationic surfactant disturbed the electrode reaction by decreasing the recorded analytical signal up to approx. 16% in the case of 200 μmol L^{-1} concentration of CTAB or Triton X-100. The suppressive influence of magnesium stearate on recorded signal was observed only for its highest tested concentration, most likely because of high amount of solid phase in measuring vessel, which impedes the electrode reaction to occur on the surface of the sensor.

Additionally, the effect of biological matrices of synthetic urine and blood serum, on recorded signal was evaluated. For that purpose, the Blood Serum and Urine Certified Reference Materials (CRM) were added in small portions to lithium chloride solution containing 50 μmol L^{-1} of AMT and the percentage difference of the signal was calculated (Fig. 4b). The measurements conducted in the presence of synthetic biological material proved that both urine and blood significantly affects the AMT oxidation peak current, with complete disappearance of the signal at 4% content of the biological matrix in the volume of the supporting electrolyte. These observations suggest that developed sensor could be used for determination of AMT in biological matrix only at high concentrations, due to high dilution rate or for samples previously subjected to extraction, in order to reduce the matrix influence.

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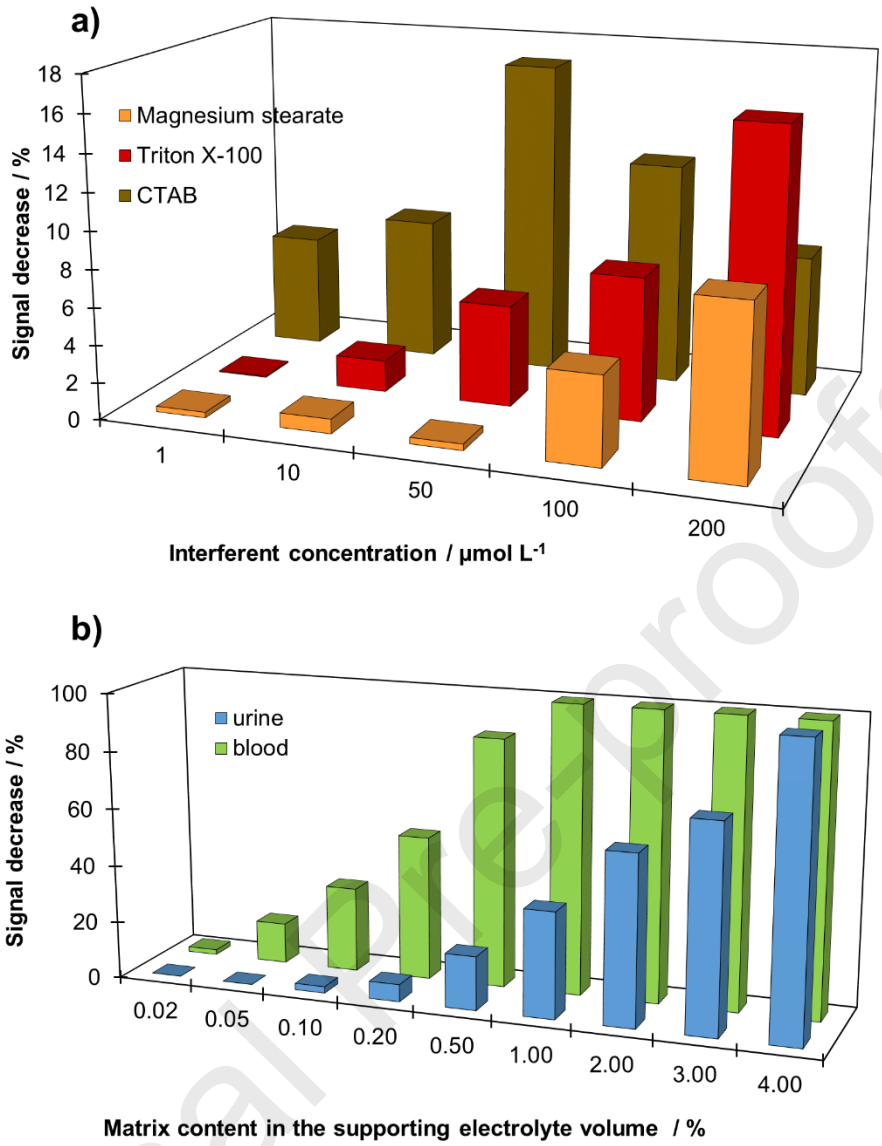


Figure 4. The signal changes caused by the presence of various compounds (a) such as magnesium stearate (orange), Triton X-100 (red) and CTAB (brown) and the occurrence of synthetic biological matrix (b) urine (blue), blood serum (green) in lithium chloride solution (0.5 mol L^{-1}) containing 50 µmol L^{-1} of AMT. DPV parameters: E_s = 7 mV; t_p = 20 ms; dE = 50 mV; t_{acc} = 180 s; E_{acc} = 0 mV.

Conducted interference studies proved that compounds and ions commonly present in pharmaceuticals and environmental waters should not affect the signal recorded for AMT using a fabricated sensor, and thus, makes it possible to apply FeZ-G/GCE for determination of AMT in tablets and water without special sample preparation. Moreover, the possible interference risk caused by the presence of high concentrations of surfactants and components of biological matrices in real samples should be minimized through application of standard addition method. In the case of the presence of both multiplicative and additive interference effects, more complex calibration approaches, like integrated calibration [52] or generalized calibration strategy [53] could be also employed.

3.7 Determination of amitriptyline in pharmaceutical, water and urine samples

To confirm the applicability of developed sensor, the Fe-zeolite/graphite modified GCE was used for AMT determination in pharmaceutical, water and urine samples. Commercially available medicament containing amitriptyline hydrochloride, previously subjected to initial preparation (see Section *2.3.4 Samples preparation*) was analyzed using standard addition method. Quantification procedure was repeated three times with the use of one sensor and the relative error (RE) with confidence intervals (with significance level 0.05) and relative standard deviation (RSD) were calculated (Table 3). The good agreement between received and declared amount of AMT in tested pharmaceutical, with RE and RSD values lower than 1.3%, suggest that application of developed sensor allows for accurate and precise determination of AMT in medicaments.

Table 3. Results of AMT determination in commercial formulation.

Sample	**Amount declared by manufacturer / *mg***	**Amount received / *mg***	**RE* / *%***	**RSD / *%***
Amitriptylinum VP	10.00	9.91 9.80 9.90	-1.3 ± 1.0	0.62

* mean value (n = 3)

Vistula water, Waste Water and Surface Water Certified Reference Materials were prepared as described in Section *2.3.4*. Due to no AMT was present in collected water or CRMs, all samples were spiked with the AMT to obtain final concentration of 20.00 μmol L^{-1}. The obtained concentrations of AMT in tested samples with corresponding recoveries are shown in Table 4. The recovery of AMT in water samples ranged from 98.0 to 99.2% with RSD lower than 1.0%. The confidence intervals achieved for recovery values were less than 2.0%, which prove good repeatability. The satisfactory precision and excellent accuracy confirm reliability of FeZ-G/GCE for AMT determination in environmental samples.

The usefulness of proposed sensor was also tested on the example of AMT quantification in sample with more complex matrix, such as synthetic urine. According to interference studies (see Fig. 4b), the synthetic urine matrix highly affect the peak current recorded for AMT, therefore urine sample was enriched to obtain 250 μmol L^{-1} of AMT and 200 μL of sample was added to 9.8 mL of lithium chloride solution, which represented 2% of the supporting electrolyte content. The determination of AMT in urine sample was performed analogously to water analysis, and based on achieved results the recovery and RSD were calculated (Table 4). For such amount of urine CRM, the AMT peak current decreased by approx. 50% in comparison to measurements conducted without presence of biological matrix. However, due to application of standard addition method, the multiplicative interference effects caused by the presence of a biological matrix did not significantly affect the accuracy and precision of performed analysis, with reasonable recovery of 95.5% and RSD of 1.84%. The achieved results prove the wide applicability of proposed method, even for samples with complex matrixes.

Table 4. Results of AMT determination in spiked water and urine samples.

Sample	Amount added / *µmol L^{-1}*	Amount received / *µmol L^{-1}*	Recovery* / *%*	RSD / *%*
Vistula water	20.00	19.55 19.80 19.44	98.0 ± 1.6	0.94
Surface water CRM	20.00	19.84 19.76 19.74	98.9 ± 0.5	0.27
Waste water CRM	20.00	19.69 20.04 19.89	99.2 ± 1.6	0.93
Synthetic urine	250.0	238.0 243.5 234.8	95.5 ± 3.0	1.84

* mean value (n = 3)

4. Conclusions

In a present work, a novel modified working electrode, based on natural zeolite – clinoptilolite, for selective and sensitive determination of tricyclic antidepressant drug amitriptyline (AMT) was introduced. The proposed sensor was fabricated by drop casting of graphite (G) powder and Fe^{3+} exchanged zeolite (FeZ) dispersed in polystyrene and acetone suspension onto glassy carbon electrode (GCE) surface. EIS and CV measurements confirmed high surface area, enhanced electrical conductivity and improved catalytic properties of fabricated sensor in comparison to bare GCE and GCE covered with graphite layer. The conducted research proved that the exceptional performance of proposed sensor is directly associated with the presence of zeolite. The FeZ-G/GCE showed remarkable short- and long-term stability, good sensitivity, wide linear range and high selectivity towards amitriptyline determination. The developed procedure of GCE surface modification with zeolite-based composite by drop casting method seems to be a good alternative to commonly used zeolite paste electrodes. It is also worth emphasizing that developed sensor stands out from the previously reported works due to possibility of conducting the analysis in a neutral environment, which promotes green analytical chemistry principles. Application of developed sensor allowed for accurate and precise determination of AMT in pharmaceutical formulation, river, surface and waste water certified reference materials and synthetic urine samples with good recovery.

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Author Statement

Maria Madej – Conceptualization, Methodology, Investigation, Validation, Formal analysis, Resources, Data curation, Visualization, Writing – Original Draft, Writing - Review & Editing

Katarzyna Fendrych – Methodology, Investigation, Validation, Formal analysis, Resources, Writing – Original Draft

Radosław Porada – Methodology, Investigation, Validation, Formal analysis, Writing – Original Draft

Mateusz Flacha – Investigation, Validation, Formal analysis

Jolanta Kochana – Conceptualization, Methodology, Data Curation, Supervision, Writing-Reviewing and Editing

Bogusław Baś – Conceptualization, Supervision, Writing – Reviewing and Editing

Highlights:

1) The Fe^{3+}-clinoptilolite-graphite modified glassy carbon electrode was fabricated.
2) The use of Fe^{3+}-zeolite promoted the electrochemical properties of the sensor.
3) The FeZ-G/GCE was applied for amitriptyline determination in various samples.
4) The sensor exhibited a high selectivity and excellent long-term stability.

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